

## Control work 3 · Roots and rational exponents

The Chapter I assessment, followed by a work-on-mistakes lesson sorted by error type.

LESSON 36–37

2 × 40 MIN

ALGEBRA 8, §§8–10

STAGE 9 · UNIT 1 CHECK

## LESSON CLOCK



Setting up 2 min

The paper 36 min

Collect in 2 min

## LEARNING OBJECTIVES

- Assess roots, rational exponents, standard form and decimals.
- Sort the errors by type rather than by question.
- Re-solve every task that was lost.

## TEXTBOOK

National Revision of §§8–10

Cambridge Learner's Book  
pp. 9–20

Workbook Workbook Unit 1

### 01 Explanation

#### Lesson 36 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each, 14 in total:

- tasks 1–2 · simplify a root (§8)
- task 3 · rationalise a denominator (§8)
- tasks 4–5 · evaluate a power with a rational exponent (§9)
- task 6 · simplify a letter expression (§10)
- task 7 · standard form or a recurring decimal (Cambridge insert)

#### Lesson 37 — work on mistakes (40 minutes)

## THE FOUR ERRORS THIS PAPER PRODUCES

1. **Splitting a root over a sum** — writing  $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ .
2. **Numerator and denominator of the index swapped** — reading  $8^{2/3}$  as  $(8^3)^2$ .
3. **A negative index treated as a negative answer** —  $4^{-1/2}$  given as  $-2$ .
4. **Standard form not tidied** — leaving  $12 \times 10^9$ .

The Hard set below is seven pieces of wrong working built from these four.

## 02 Worked examples

**EXAMPLE 1** Find the error:  $\sqrt{16 + 9} = 4 + 3 = 7$

- ① A root does not split over a sum.  
Only over a product or a quotient.

- ②  $16 + 9 = 25$   
Do the addition first.

- ③  $\sqrt{25} = 5$

ANSWER

5

**EXAMPLE 2** Find the error:  $4^{-1/2} = -2$

- ① A negative index means a reciprocal, not a negative number.

- ②  $4^{-1/2} = \frac{1}{4^{1/2}}$

- ③  $= \frac{1}{2}$

ANSWER

$\frac{1}{2}$

## 03 Interactive model

Show each piece of wrong working, take a vote on the error type, then reveal.

**Diagnose the error**

**Claimed:**  $\sqrt{16 + 9} = 7$

- ① The product rule is  
 . There is no sum rule.

$$\begin{aligned} \sqrt{ab} \\ = \\ \sqrt{a} \\ \cdot \\ \sqrt{b} \end{aligned}$$

- ②  $16 + 9 = 25$

, so  $\sqrt{25}$   
 the  
 root  
 is

- ③  $= 5$

, One counter-example is enough to kill the false rule.  
 not  
 7.

**04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH          | O'ZBEKCHA              | РУССКИЙ             |
|------------------|------------------------|---------------------|
| Control work     | Nazorat ishi           | Контрольная работа  |
| Work on mistakes | Xatolar ustida ishlash | Работа над ошибками |
| Root             | Ildiz                  | Корень              |
| Exponent         | Ko'rsatkich            | Показатель          |
| Standard form    | Standart ko'rinish     | Стандартный вид     |
| Evaluate         | Hisoblash              | Вычислить           |
| Justify          | Asoslash               | Обосновать          |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$\sqrt{25 + 144}$  equals:

17

13

169

5 + 12

$8^{2/3}$  equals:

4

64

16

262144

$9^{-1/2}$  equals:

-3

$\frac{1}{3}$

3

$-\frac{1}{3}$

$20 \times 10^9$  in standard form is:

$$20 \times 10^9$$

$$2 \times 10^{10}$$

$$2 \times 10^8$$

$$0.2 \times 10^{11}$$

## 06 Practice · 21 problems

---

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Simplify  $\sqrt{48}$

ANSWER  $4\sqrt{3}$

2. **Task 2.** Simplify  $\sqrt{6} \cdot \sqrt{24}$

ANSWER 12

3. **Task 3.** Rationalise  $\frac{4}{\sqrt{2}}$

ANSWER  $2\sqrt{2}$

4. **Task 4.** Evaluate  $16^{1/2}$

ANSWER 4

5. **Task 5.** Evaluate  $27^{2/3}$

ANSWER 9

6. **Task 6.** Simplify  $a^{1/2} \cdot a^{3/2}$

ANSWER  $a^2$

7. **Task 7.** Write 0.0035 in standard form

ANSWER  $3.5 \times 10^{-3}$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Simplify  $\sqrt{75}$

ANSWER  $5\sqrt{3}$

2. **Task 2.** Simplify  $\sqrt{8} \cdot \sqrt{18}$

ANSWER 12

3. **Task 3.** Rationalise  $\frac{9}{\sqrt{3}}$

ANSWER  $3\sqrt{3}$

4. **Task 4.** Evaluate  $64^{1/3}$

ANSWER 4

5. **Task 5.** Evaluate  $25^{-1/2}$

ANSWER  $\frac{1}{5}$

6. **Task 6.** Simplify  $\frac{b^{5/4}}{b^{1/4}}$

ANSWER  $b$

7. **Task 7.** Write  $\frac{2}{3}$  as a decimal

ANSWER 0.(6)

Hard · stretch and reason

7 PROBLEMS



1. Find the error:  $\sqrt{16 + 9} = 7$

ANSWER Roots do not split over a sum. Correct: 5.

2. Find the error:  $8^{2/3} = (8^3)^2$

ANSWER Index upside down. Correct: 4.

3. Find the error:  $4^{-1/2} = -2$

ANSWER A negative index gives a reciprocal. Correct:  $\frac{1}{2}$ .

4. Find the error:  $(4 \times 10^5)(5 \times 10^4) = 20 \times 10^9$

ANSWER Not tidied. Correct:  $2 \times 10^{10}$ .

5. Find the error:  $\sqrt{a^2} = a$  for every  $a$

ANSWER True only for  $a \geq 0$ ; for  $a < 0$  it is  $-a$ .

6. Find the error:  $\frac{1}{\sqrt{2}} = \sqrt{2}$

ANSWER Rationalising gives  $\frac{\sqrt{2}}{2}$ , which is about 0.71, not 1.41.

7. Find the error:  $0.(3) = \frac{3}{10}$

ANSWER  $\frac{3}{10} = 0.3$  exactly, not recurring. Correct:  $\frac{1}{3}$ .

## 07 Homework

### After the work-on-mistakes lesson

Re-solve every task you lost marks on, then these three.

1. Simplify  $\sqrt{147}$  and rationalise  $\frac{5}{\sqrt{5}}$
2. Evaluate  $32^{3/5}$  and  $49^{-1/2}$

3. Write out the four errors from this lesson in your own words with one example each.